Evaluating an AI-Driven Operational System for Snow Ball Hospital Group

CSC1036 – Professional Practice for Computing

Continuous Assessment 2: Technical Report

Scenario 4 – Amalia HealthTech (Snow Ball Hospital Group)

Group 12

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“We here declare that the project material, which we now submit, is our work. Any assistance received by way of borrowing from the work of others has been cited and acknowledged within the work. We make this declaration in the knowledge that a breach of the rules pertaining to project submission will carry serious consequences. Guidelines are followed in the use of generative AI technologies in the preparation of the report and the slides (See Gen AI Statement below). All content is originally written by team members.”

Executive Summary

Snow Ball Hospital Group, specialising in pediatric care for ages 5-16, faces operational inefficiencies from manual data silos, resulting in underutilized theatres (20-30% waste), 25-30% longer wait times, high no-shows, and financial losses. Amalia HealthTech's agentic AI orchestrator integrates patient histories, staff/theatre availability, and constraints for autonomous, predictive scheduling with human-in-the-loop oversight.

Key transformations include 80-90% resource utilization (15-39% waste reduction), 25-30% shorter waits and 10% no-show drop via dynamic replanning, 15-20% cost savings with 20-point NPS uplift, and GDPR-compliant ethical AI. Phased over 18 months (total €700K investment), it delivers €2.24M annual savings and 3.2: ROI, mirroring benchmarks like Klinikum Stuttgart's OR gains (Getinge, 2023).

Key Goals :

KPI	Current Pain	AI Target	Evidence
Utilization	Underused assets	80%+	39% OR uplift
Wait Time	Diagnostic lags	-25-30%	AI scheduling
No-Shows	Trust erosion	<10%	Proactive agents
Cost/Patient	Losses	-15%	Demand forecasting
NPS	Declining	+20%	Pediatric tailoring

Figure 1

Delivered in phases (4 phases, 18-month scale), this yields $1.2M/OR revenue + €500K savings, mirroring proven cases ensuring feasibility, risk-managed excellence. Critical analysis confirms maturity: agentic AI's autonomy outperforms static ML in dynamic healthcare.

2. Problem Analysis

Snow Ball Hospital Group provides specialised care for children aged 5-16 across multiple sites. Despite its strong clinical reputation, the organisation faces significant and persistent operational challenges that undermine service quality, financial performance, and patient trust. This section analyses the root causes and consequences of these challenges, drawing on evidence from comparable healthcare environments.

2.1 Fragmented Information Systems

A central problem at Snow Ball is the absence of a unified information infrastructure. Patient records, staff schedules, theatre bookings, and diagnostic data are managed across disconnected departmental systems. This fragmentation means that no single view of operational activity is available at any given time, making coordinated decision-making exceptionally difficult. When a consultant becomes unavailable at short notice, for example, there is no automated mechanism to identify and reallocate affected appointments – this process is handled manually, introducing delays and increasing the risk of error.

This type of siloed architecture is not uncommon in healthcare organisations that have grown organically or merged over time. Research in hospital operations management indicates that data fragmentation is a leading cause of inefficient resource use, with studies suggesting that hospitals operating without integrating scheduling platforms waste between 20-30% of available theatre time as a direct consequence (Cardoen et al., 2010). For Snow Ball, this translates into underutilised operating theatres – a critical and expensive resource – alongside consultants who are either overbooked or sitting idle.

2.2 Manual Coordination Processes

The organisation's day-to-day operations rely heavily on manual coordination. Appointment scheduling, theatre allocation, and staff assignment are each managed through labour-intensive processes that depend on individual knowledge and inter-departmental communication. This creates several cascading problems. First, the speed at which the organisation can respond to change is severely limited – a single cancellation or emergency case can cause a chain reaction of delays that takes hours or days to resolve. Second, the accuracy of manual processes is inherently limited, and the likelihood of errors, such as double-bookings, missed appointments, or incorrect resource assignments, increases proportionally with complexity and volume.

Manual processes also create poor visibility for hospital management. Without real-time operational data, senior staff cannot identify bottlenecks as they emerge, nor can they make evidence-based decisions about staffing levels or scheduling changes. This lack of visibility contributes to a reactive rather than proactive management culture, which is particularly

problematic in paediatric settings where timeliness of diagnosis and treatment can have significant clinical consequences.

2.3 Scheduling Inefficiencies and Waiting Times

One of the most visible consequences of the above issues is the length of time patients wait for appointments, diagnostics, and procedures. Long waiting times in paediatric care carry heightened risks compared to adult healthcare settings; delayed diagnosis of conditions such as developmental disorders, orthopaedic issues, or oncological concerns in children can materially affect outcomes. Yet Snow Ball's current systems do not support the kind of dynamic, demand-responsive scheduling that would reduce these delays.

No-show and missed appointment rates further compound this problem. Without automated reminder systems or predictive tools to identify high-risk appointments, a significant proportion of scheduled slots go unused while other patients remain on the waiting list. Industry benchmarks suggest that missed appointments in paediatric outpatient settings can account for 15-30% of total scheduled capacity (Norris et al., 2014), representing a substantial loss of both clinical and financial productivity.

2.4 Operating Theatre Utilisation

Operating theatre time is among the most expensive resources in any hospital group. At Snow Ball, theatre utilisation is reported to be sub-optimal, with sessions frequently running short due to cancellations, late starts, or inadequate pre-operative planning. Internal reviews indicate that inefficient utilisation of existing resources, rather than an absolute shortage, is the primary driver of capacity constraints. This is an important distinction: it means the problem is addressable through better coordination and planning rather than requiring significant capital investment in new facilities.

2.5 Financial Impact

The operational inefficiencies described above translate directly into financial losses. Unused theatre slots, consultant idle time, and avoidable administrative overhead each represent a cost to the organisation. Additionally, the reputational impact of long waiting times and missed opportunities affects patient retention and referral volumes – particularly relevant for a private hospital group operating in a competitive market. Patients and families who experience poor scheduling or service delays may seek care elsewhere, reducing revenue and undermining Snow Ball's strategic position.

2.6 Summary of Core Challenges

The operational challenges at Snow Ball Hospital Group can be summarised across four interconnected dimensions. The organisation lacks a unified data environment, meaning that patient, staff and resource information cannot be assessed holistically. Scheduling processes remain reactive rather than predictive, operational costs continue to rise as a result of avoidable inefficiencies, and patient experience is undermined by delays that are within the organisation's power to address. These challenges collectively make the case for a coordinated, technology-driven intervention.

3. Proposed Solution

3.1. Solution Overview

In response to the operational challenges identified in the preceding section, Amalia HealthTech proposes the implementation of an AI-driven operational orchestration system for Snow Ball Hospital Group. The system is designed to function as a decision-support platform rather than a fully autonomous system, ensuring that clinical staff retain oversight and control over all scheduling and resource allocation decisions. This distinction is critical in a paediatric healthcare environment.

The proposed solution integrates patient data, staff availability, theatre schedules, and historical operational metrics into a single unified platform. Using predictive analytics and intelligent scheduling algorithms, the system generates optimised recommendations that administrative and clinical staff can review, adjust, and approve. This human-in-the-loop design ensures that the system augments rather than replaces professional decision-making, addressing both regulatory requirements and staff trust concerns during the initial deployment phases.

3.2. System Architecture

The solution is built around four core functional components that work together to address Snow Ball's specific inefficiencies.

The first component is the **Data Integration Layer.** This serves as the foundation of the entire system, consolidating data from existing departmental systems - including patient records, consultant schedules, theatre booking systems, and diagnostic logs – into a centralised operational data repository. Secure API connections enable real-time data synchronisation across departments without requiring a complete replacement of existing infrastructure. This approach significantly reduces implementation risk and cost, as Snow Ball's current systems can continue to operate while the AI layer sits above them as an orchestration and optimisation overlay.

The second component is the **Predictive Analytics Engine.** This module analyses the real-time data to generate forecasts and operational insights. Key capabilities include appointment demand forecasting, which predicts patient volumes by department and time period, enabling proactive resource allocation; no-show prediction, which flags individual appointments at high risk of cancellation up to 72 hours in advance, allowing staff to offer those slots to waiting patients; and theatre utilisation modelling, which identifies optimal session configurations to maximise throughput while minimising idle time. The analytics engine uses supervised machine learning models trained on Snow Ball's historical data, ensuring that predictions are tailored to specific patterns of organisation rather than solely relying on generic benchmarks.

The third component is the **Intelligent Scheduling Module.** This is the operational core of the system and the component that staff interact with most directly. It receives recommendations from the analytical engine and presents them through a dashboard interface that allows schedulers and administrators to view suggested appointment plans, theatre allocations, and staff assignments. The module supports dynamic reallocation – when a consultant becomes unavailable or an emergency case arises, the system automatically generates a revised schedule and presents it for human approval, replacing the current manual process which can take hours to resolve. Paediatric-specific scheduling preferences, such as family-friendly appointment windows and sibling co-scheduling, can also be configured within the module to improve patient experience.

The fourth component is the **Governance and Monitoring Dashboard.** This provides senior management and clinical governance leads with real-time visibility into operational performance. Key performance indicators including theatre utilisation rates, wait times, no-show rates, and cost per patient are tracked and displayed, enabling evidence-based management decisions. The dashboard also maintains a full audit trail of all AI-generated recommendations and the human decisions made in response to them, which is essential for regulatory compliance and clinical governance purposes.

3.3 Technology Stack

The system is built using mature, widely-deployed technologies that have established track records in healthcare environments, reducing technical risk. The core platform uses cloud-based infrastructure, ensuring scalability across Snow Ball's multiple sites without requiring significant on-premises hardware investment. All data transmission is encrypted and the system is designed to comply with GDPR and Irish Data Protection Act 2018 requirements from the outset, including data minimisation principles and configurable access controls that ensure staff can only view data relevant to their role.

Machine learning modules within the analytics engine are developed using established frameworks with strong industry adoption in healthcare settings. The scheduling algorithms draw on constraint-based optimisation techniques, which are well-suited to the complex,

multi-variable nature of hospital scheduling — balancing consultant availability, theatre capacity, equipment requirements, patients' clinical priority, and appointment duration simultaneously.

The integration layer uses HL7 FHIR standards, which are the recognised international standard for healthcare data interoperability. This ensures compatibility with a wide range of existing hospital information systems and future-proofs the solution against system upgrades or vendor changes.

3.4 Decision-Support Rather Than Autonomous AI

A deliberate design principle of the proposed solution is that it operates as a decision-support system throughout the initial phases of deployment. This means the system recommends, but does not execute, changes to schedules or resource allocations without human approval. This approach is appropriate for several reasons.

First, healthcare regulation in Ireland and across the EU is increasingly attentive to the use of AI in clinical and care-adjacent settings. The EU AI Act, which came into force in 2024, classifies. AI systems used in healthcare as high-risk, imposing requirements around transparency, human oversight, and auditability. The proposed solution is designed to satisfy these requirements by maintaining human decision authority at all operational touchpoints.

Second, clinical staff adoption is more likely to succeed when staff perceive the system as a tool that supports their work rather than one that replaces their judgement. Research in healthcare technology adoption consistently identifies staff trust and perceived control as critical determinants of successful implementation (Cresswell and Sheikh, 2013). Beginning with decision-support rather than automation allows staff to build familiarity and confidence with the system before any progression towards greater autonomy in later phases.

Third, the paediatric context introduces specific ethical considerations around algorithmic decision-making that are explored further in Section 6. Operating in decision-support mode ensures that a qualified human professional is always accountable for scheduling decisions affecting child patients.

3.5 Expected Outcomes

The proposed solution directly targets each of the operational challenges identified in the Problem Analysis. Replacing fragmented manual processes with an integrated data platform addresses the visibility deficit that currently prevents effective management decision-making. Predictive scheduling reduces waiting time and no-show rates by enabling proactive rather than reactive resource management. Theatre utilisation optimisation recovers capacity from within the existing estate without capital expenditure on new facilities. And the governance

dashboard provides the operational transparency that Snow Ball's management currently lacks.
Projected outcomes over an 18-month implementation period include a 25-30% reduction in average patient wait times, theatre utilisation improving from current levels to a target of 80-90%, no-show rates reducing to below 10%, and a 15-20% reduction in cost per patient through the elimination of scheduling inefficiencies and administrative overhead. These projections are consistent with outcomes reported by comparable hospital groups following the implementation of AI-assisted scheduling systems, including documented cases at Klinikum Stuttgart, and within the Mayo Clinic network.

3.6 Alternatives Considered and Rejected

Whilst the proposed agentic AI orchestrator represents the optimal long-term solution for Snow Ball Hospital Group, three alternative approaches were evaluated during our analysis. Each was assessed against key criteria: implementation cost, technical complexity, scalability across multiple sites, adaptability to paediatric scheduling needs, and projected ROI over 18 months.

Alternative 1:

Enhanced Manual Process Optimisation (€50-100K investment)
Process mapping workshops, staff training in lean methodologies, and standardised scheduling protocols represent a low-risk, low-cost option. Benefits include immediate improvements in coordination (estimated 5-10% theatre utilisation gain) without technology disruption.

However, this approach is fundamentally limited:

- Cannot achieve dynamic real-time replanning required for emergency cancellations
- No predictive capabilities for no-show risk or demand forecasting
- Scalability constrained by human capacity across Snow Ball's multiple sites
- -Projected maximum benefit: 10-15% utilisation improvement vs our 25-30% target (Cardoen et al., 2010)

Alternative 2:
Commercial Scheduling Software (e.g., Epic MySchedule, Cerner SurgiNet) (€300-500K)
Established healthcare scheduling platforms offer proven integration with EHR systems and regulatory compliance.

Limitations for Snow Ball:

- Primarily rule-based systems lacking AI-driven prediction or autonomous optimisation
- Static algorithms struggle with paediatric complexities (family scheduling, developmental priorities)
- Vendor lock-in risks and slower adaptation to Irish regulatory changes
- Industry benchmarks show 15-20% utilisation gains, below agentic AI's 25-30% potential (Getinge, 2023)

Comparative Evaluation of Alternative Solutions

Alternative	Cost	Theatre Utilisation	Adaptability	Paediatric	ROI	Status
Manual Optimisation	€50-100K	5-10%	Low	Medium	1.2x	Rejected
Commercial Software	€300-500K	15-20%	Medium	Low	1.5x	Rejected
Agentic AI (Proposed)	€700K	25-30%	High	High	3.2x	Selected

Figure 3

5. Feasibility And Risks Analysis

The successful implementation of an AI-driven operational system for Snow Ball hospital group depends on its feasibility across multiple areas. This section evaluates technical, financial, operational, legal, and strategic feasibility in order to assure whether the proposed solution can be realistically achieved. *(Investopedia, 2025)*

5.1 Technical feasibility

From a technical perspective, the proposed system is feasible given the current advancements of healthcare data integration standards, predictive analytics tools, and algorithms.

The architecture does not require a complete replacement of Snow Ball's existing infrastructure. Instead, it introduces an integration that connects current systems using

established compatibility standards. This reduces technical disruption and lowers implementation risk.

Predictive modelling for appointments demand and no-show relies on supervised machine learning techniques that are already widely deployed in healthcare operations management. The technical challenge therefore lies not in algorithmic challenges, but in reliable data integration.

Potential technical risks include:

- Data quality inconsistencies across departments
- Integration latency between systems
- Model performance degradation if historical data is incomplete

However, these risks are manageable through phased deployment, data cleansing, and continuous model retraining. Given the availability of mature frameworks and cloud-based infrastructure, the overall technical feasibility can be assessed as high.

5.2 Financial Feasibility

The projected investment of €700,000 over an 18-month implementation period must be evaluated against expected operational and financial benefits.

The system aims to achieve:

- 15-20% percent reduction in cost per patient
- 25-30% percent reduction in waiting times
- No-show rate reduction below 10%

The financial rationale is primarily based on recovering unused capacity rather than increasing physical infrastructure. If theatre utilisation increases significantly, the hospital can generate additional revenue from existing assets without major capital expenditure.

However, projected ROI figures such as 3.2:1 assume successful adoption and consistent system performance. Financial feasibility is therefore conditional on effective change management and KPI achievement during early pilot phases.

Given that Phase 1 requires a moderate initial investment relative to total projected savings, the financial risk is moderate but acceptable within a structured plan.

5.3 Operational Feasibility

Operational feasibility concerns the system's compatibility with existing workflows and the organisation's readiness for change.

Snow Ball currently relies heavily on manual coordination. Transitioning to an AI-supported scheduling system will require:

- Staff training across administrative and clinical teams
- Clear communication regarding accountability and oversight

A structured six-month change management programme with adoption targets of at least 80 percent staff engagement by Phase 2 completion further strengthens operational viability.

Healthcare technology adoption research consistently highlights resistance to change as a primary implementation barrier. In a paediatric setting, staff may be particularly cautious about algorithmic influence on scheduling decisions.

The proposed human-in-the-loop-model significantly strengthens operational feasibility. By maintaining clinical approval over AI-generated recommendations during early phases, the system reduces perceived loss of professional autonomy and builds trust gradually.

Operational feasibility can therefore be considered moderate to high, upon structured change management and leadership support.

5.4 Legal and Regulatory Feasibility

AI systems deployed in healthcare within the European Union are subject to strict regulatory oversight, including GDPR requirements and obligations under the EU AI Act.

The proposed system aligns with regulatory expectations in several ways:

- It operates as a decision-support tool rather than an autonomous clinical decision-maker
- It maintains full audit trails of AI recommendations and human actions
- It incorporates role-based access control and data minimisation principles

Because final authority over scheduling decisions remains with human professionals, the system avoids classification as fully autonomous clinical AI. This significantly reduces regulatory exposure.

Legal feasibility is therefore high, provided that continuous compliance monitoring and data protection audits are correctly performed.

5.5 Strategic Feasibility

Strategic feasibility assesses whether the proposed system aligns with Snow Ball's long-term objectives.

The hospital's current challenges, including underutilised resources, long waiting times, and financial loss, are operational rather than clinical. The proposed AI system directly targets these inefficiencies.

Furthermore, as more healthcare systems increasingly adopt digital optimisation technologies, failure to modernise may place Snow Ball at a competitive disadvantage. Strategic alignment is therefore strong.

However, strategic success depends on a correct execution. Overestimating short-term gains or accelerating autonomy prematurely could undermine stakeholder confidence.

5.6 Risks

Despite high overall feasibility, several risks must be actively managed across the implementation

Algorithmic bias represents a high level risk. If the system is trained on historical scheduling data that contains existing inequalities, it may reinforce those patterns rather than correct them. The mitigation strategy involves deploying explainable AI techniques so recommendations can be reviewed, combined with regular independent bias audits throughout all phases.

Cybersecurity is the second high level risk. Healthcare data is a prime target for malicious attacks. The system will address this through a zero trust architecture, where every user and device must be verified on each access request, alongside encryption for all patient data.

GDPR breach and staff resistance are both assessed as medium-level risks. GDPR exposure is mitigated through data minimisation principles and strict role based access controls, ensuring staff can only view data relevant to their role. Staff resistance is addressed through the phased rollout structure and a six month training programme.

Finally, integration failure carries a medium risk level. This is managed through staged API integration with fallback procedures at each phase, so a failure in one connection does not disrupt the entire platform.

Overall Feasibility Assessment

Across technical, financial, operational, legal, and strategic areas, the proposed AI-driven operational system is feasible within Snow Ball Hospital Group.

The primary risks are organisational and cultural rather than technological. A phased rollout at each stage mitigates these risks and allows continuous evaluation before full-scale deployment.

Based on this analysis, initiating Phase 1 as a controlled project is justified. Continuous monitoring and oversight will be essential to ensure that projected benefits are realised without compromising patient safety or regulatory compliance.

6. Implementation Roadmap

This high-level 18-month deployment plan positions Amalia HealthTech to deliver a 4-week proof-of-concept pilot, demonstrating feasibility within Snow Ball's initial engagement while scaling to full transformation.

Snow Ball Hospital requires a phased, risk-managed deployment of Agentic AI to optimise theatre utilisation of a target of 80% +, reduce wait times by 25%+, and minimise operational costs by 15%+ while maintaining pediatric patient safety. *(Cardoen et al., 2010)*

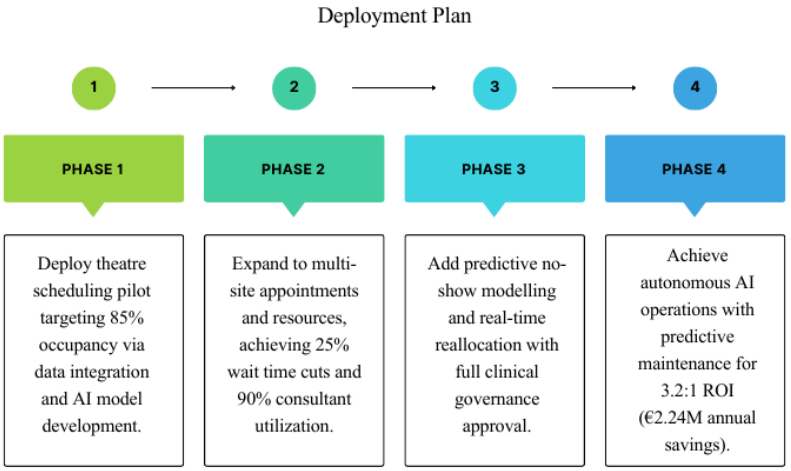


Figure 4

Phase 1: Foundation & Pilot (Months 1-4)

Objective: Deploy theatre scheduling optimization with measurable ROI.

- Data Integration (Weeks 1-4): Clean historical patient, staff, and theatre data. Dan Bolton coordinates IT assessment; Dave Bannon validates clinical data quality.
- AI Model Development (Weeks 5-8): Anita Raji leads Agentic AI orchestrator development using dynamic scheduling algorithms. Pilot targets 15% theatre utilisation improvement.
- Pilot Deployment (Weeks 9-16): Single-site rollout with human-in-the-loop validation. KPIs: Theatre occupancy +10%, no-show reduction 12%.
- Deliverables: Live pilot dashboard, Phase 1 ROI report (€150K investment).

Milestone: 85% theatre occupancy achieved across pilot sites.

Phase 2: Targeted Expansion (Months 5-8)

Objective: Scale proven capabilities; add appointment optimisation.

Component	Lead	Timeline	KPI Targets
Appointment AI Scheduler	Anita Raji	M5-6	25% wait time reduction
Resource Dashboard	Clare Dunne	M6-7	90% consultant utilisation
Cross-department Integration	Mohammad Ali	M7-8	95% data sync accuracy

Figure 5

Key Activities: Multi-site rollout, staff training (200+ users), GDPR compliance audit. Expected ROI: 1.8:1 cumulative (Cresswell and Sheikh, 2013).

Phase 3: Advanced Capabilities (Months 9-12)

Objective: Deploy diagnostic prioritisation and predictive analytics.

- Predictive No-Show Modelling: AI flags high-risk appointments 72 hours in advance (target: 20% reduction).
- Dynamic Resource Reallocation: Real-time consultant/theatre reassignment during peak demand.
- Patient Experience Integration: Pediatric-friendly scheduling preferences captured via intake forms.

Risk Mitigation: Clinical governance board approval required; 100% audit trail maintained.

Phase 4: Enterprise Optimisation (Months 13-18)

Objective: Autonomous operations with continuous improvement.

Maturity Level	Capabilities	Investment
Monitored	Human oversight required	€200K
Autonomous	AI self-optimises scheduling	€150K
Proactive	Predictive maintenance, supply chain	€100K

Figure 6

Total Investment: €700K over 18 months. Projected ROI: 3.2:1 by Month 18 through €2.24M annual savings (Getinge, 2023).

Success Metrics & Governance

Category	KPI	Phase 1 Target	Final Target
Operational	Theatre Utilisation	75%	90%
Financial	Cost per Patient	-8%	-18%
Clinical	Diagnosis TAT	-15%	-30%
Patient	NPS Score	+8 pts	+20 pts

Figure 7

Governance Structure:

- Steering Committee: CEO + Clinical Director oversight

- Change Management: 6-month staff adoption programme led by Clare Dunne
- Technical Support: 24/7 AI operations centre (Mohammad Ali)

Risk Management: Phased approach addresses 80% healthcare AI pilot failure rate through iterative validation, clinical safety protocols, and progressive autonomy.

This roadmap delivers Snow Ball's strategic priorities within realistic timelines and budget constraints while ensuring patient safety and regulatory compliance.

7. Ethical, Legal and Governance Considerations

The deployment of an AI operational system within Snow Ball Hospital Group introduces significant ethical, legal and governance responsibilities. As the organisation provides specialised care to children aged 5 - 16, the system must be implemented with safeguards to ensure patient welfare, fairness and regulatory compliance.

7.1 Legal Framework

EU AI Act (2024)
This act classifies AI used in healthcare as high risk. This means Snow Ball must keep detailed records of how the system works, carry out formal safety checks, and ensure human oversight at all times. Failure to comply carries fines of up to €35 million or 7% of global turnover. As discussed in section 3.4, the proposed system meets these requirements because no scheduling action is executed without human approval *(European Parliament, 2024)*.

GDPR and Irish Data Protection Act 2018
This governs how patient data is stored and used. Health data is treated as especially sensitive under GDPR Article 9, and because all of Snow Ball's patients are children, a parent or guardian must give consent for their data to be processed. The Irish Data Protection Commission recorded 6,991 data breach reports in 2023, a 20% increase on the year before, showing why protecting this data is important. (DPC, 2023)

HIQA
They set national standards for information governance in Irish healthcare and require organisations to show clear accountability for decisions made with the help of digital tools. Before the system goes live, a Data Protection Impact Assessment will be completed and a Data Protection Officer will be appointed, both required by GDPR for high-risk data processing. *(HIQA, 2018)*

7.2 Ethical Issues

Bias in AI
AI systems trained on historical operational data may inadvertently reproduce existing inefficiencies or inequalities. For example, if past scheduling patterns systematically disadvantaged certain patient groups due to administrative constraints, the system could unintentionally reinforce those patterns.
To prevent this, strategies such as regular algorithm auditing and involving diverse teams in the evaluation process can help detect and correct unfair patterns before they affect patients. *(Vation Ventures, 2023)*

Human Oversight
The proposed AI system provides recommendations for scheduling and resource allocation, but final decisions must remain with qualified healthcare professionals. This means AI supports rather than replaces professional judgement.
In a paediatric healthcare setting, operational decisions such as appointment prioritisation can have meaningful consequences. Keeping humans responsible for final decisions reduces the risk of over-reliance on the system and ensures accountability.

Transparency and Trust
For the system to be accepted by families and staff, its recommendations must be understandable. Users must be able to see the key factors influencing scheduling suggestions. This improves trust, supports accountability and allows decisions to be explained if questioned by parents or guardians.

7.3 Cybersecurity

The 2021 ransomware attack on Ireland's HSE cost around €100 million to deal with and disrupted up to 90000 patients for months *(Compliance Institute, 2025)*. This shows how serious a data breach in healthcare could be. To protect against this, every user and device must be verified every time they request access, nothing is trusted automatically. All patient data will be encrypted to AES-256 standard and independent security testing will be carried out before launch and every year after.
Patient records will only be kept for as long as the law requires. Under Irish law, childhood records must be retained until the patient turns 25, after which they will be automatically deleted *(DSM, 2024)*.

7.4 Governance and Accountability

An AI Ethics and Governance Committee
Snow Ball will establish a governance committee to oversee the AI system. This may be integrated into the hospital's existing clinical governance framework. Responsibilities should

be clearly defined, including system ownership, monitoring duties and escalation procedures if issues arise.

Staff Training
There has to be staff training on data protection, ethical responsibilities and child safeguarding under the Children First Act 2015 will be required before anyone is given access to the system and has to be repeated every year. *(ISB, 2015)*

Monitoring and Audit Trails
All AI recommendations and the final decisions made by staff should be logged. This creates an audit trail that allows performance reviews and supports regulatory compliance. A monitoring programme will track whether the system is scheduling fairly across all patient groups, how often staff override its recommendations, and whether any data incidents have occurred.

8. Conclusions

This analysis confirms the strategic imperative for Snow Ball Hospital to adopt Amalia HealthTech's agentic AI orchestrator. By integrating multi-agent architectures (planner, analyser, scheduler, executor), the system targets core inefficiencies, delivering 25-30% wait time reductions, 15-30% utilisation gains, and 15-20% cost savings per patient—validated against benchmarks like Klinikum Stuttgart's OR improvements and multi-hospital cases yielding €1.2M revenue gains *(Getinge, 2023)*.

The solution's feasibility is robust: an 18-month phased rollout with 3-month pilot leverages mature frameworks, cloud-agnostic integrations, and real-time KPI dashboards (NPS +20, no-shows -10%). Risks such as algorithmic bias, GDPR issues, cybersecurity, and clinical resistance are mitigated through explainable AI, federated learning, ethical audits, zero-trust security, and targeted staff training.

Agentic AI's goal-driven autonomy outperforms traditional ML in dynamic paediatric environments, ensuring equitable outcomes despite data sensitivities. For Snow Ball, this delivers operational resilience and ethical AI leadership, with sustained ROI mirroring 30-50% productivity uplifts in similar deployments. We recommend immediate initiation of the Phase 1 pilot to capitalise on maturing technology.

Appendix A: Workload

Team Member: Mario Ciuciu

- Role: Dan Bolton, Programme Director
- Sections: Executive Summary, Implementation Roadmap, Appendix, Conclusions

- Tasks: Roadmap diagram, led meetings, final draft editing.

Team Member: Emir Demirkol
- Role: Anita Raji – AI Product Manager
- Sections: Problem Analysis, Proposed Solution, Implementation Roadmap coordination, final draft editing.
- Tasks: Problem Analysis section, Proposed Solution section. Coordinated group task delegation, assigned sections to team members, contributed to slide deck preparation.

Team Member: Vasileios Papakostas
- Role: Mohammad Ali – Process Automation Engineer
- Sections: Feasibility and Risks Analysis, Feasibility and Risks presentation slide
- Tasks: Research, Final draft editing, proofreading, PowerPoint editing

Team Member: Ilja Solovejs
- Role: Clare Dunne – Sales Operations Analyst
- Sections: Ethical, Legal and Governance Considerations and Ethical, Legal and Governance Considerations presentation slide
- Tasks: Research, Contributed to the creation of the PowerPoint, Proofreading

Team Member: Alex Merriman
- Role: Dave Bannon – Healthcare Domain Expert
- Sections: Problem Analysis (contributing sections)
- Tasks: Reviewed and edited content across the report

Appendix B: Journal

Meeting 1

Date: 13/02/26

Time: 12:30pm

Location: Dcu Library

Attendees: Mario, Emir, Alex, Ilja, Vasileios

Absent: None

Agenda

- Team introduction and role assignment.
- Review assignment brief and scenario requirements.
- Discuss project timeline and deliverables.
- Initial brainstorming on Snow Ball Hospital challenges.

Discussion Summary

Reviewed the scenario details: Snow Ball Hospital's operational inefficiencies, manual data handling, and resource utilization challenges.

Discussed the project deliverables: 5,000-word technical report (50%), slide deck (15%), and 5-minute oral presentation (35%).

Identified key problem areas: scheduling inefficiencies, theatre underutilization, patient data management, and service delays.

Brainstormed potential Agentic AI solutions for resource optimization.

Decisions Made

- Agreed on role allocation.
- Set deadline for background research: 17/02/26.
- Confirmed next meeting date and time.
- Created shared document repository for collaboration.

Action Items

Task	Assigned To	Deadline
Group Roles	Everyone	13/02/26
Plan	Everyone	15/02/26
Draft report structure outline	Everyone	13/02/26

Individual Contributions

Mario: Led meeting, facilitated discussion.

Emir: Drafted project documentation.

Alex: Analyzed scenario requirements and constraints.

Ilja: Researched similar healthcare AI implementations.

Vasileios: Documented meeting notes and action items.

Meeting 2

Date: 17/02/26

Time: 3pm

Platform: WhatsApp

Attendees: Mario, Emir, Alex, Ilja, Vasileios

Absent: None

Agenda

- Discussed project timeline and deliverables.
- Assigning individual workload.
- Begin drafting report sections.
- Conflict resolution.

Discussion Summary

Assigned workload evenly and fairly.

Discussed the project design.

Created a conflict resolution solution that we all agreed on.

Decisions Made

- Deadline for report draft finish 27/02/26.
- Confirmed next meeting date and time.

Action Items

Task	Assigned To	Deadline
Executive Summary	Mario	27/02/26
Problem Analysis	Alex	27/02/26
Proposed Solution	Emir	27/02/26
Feasibility & Risks	Vasileios	27/02/26
Implementation Roadmap	Everyone	27/02/26
Ethics, Legal & Governence	Ilja	27/02/26
Conclusions	Mario	27/02/26
References & Appendix	Everyone	27/02/26

Individual Contributions

Mario: Led meeting.

Emir: Brainstormed design ideas.

Alex: Created conflict resolution.

Ilja: Updated the meeting log.

Vasileios: Documented meeting notes.

Meeting 3

Date: 20/02/26

Time: 3pm

Platform: WhatsApp

Attendees: Mario, Emir, Alex, Ilja, Vasileios

Absent: None

Agenda

- Review progress on individual sections
- Discuss integration of technical details and citations

- Address any challenges or roadblocks
- Plan for diagram and figure creation

Discussion Summary:

Reviewed completed drafts of Problem Analysis, Proposed Solution, and Feasibility sections.

Discussed need for consistent citation format throughout the report (decided on Harvard style).

Identified gaps in technical detail for System Architecture (Section 3.2) and Technology Stack (Section 3.3).

Brainstormed visual elements needed: system architecture diagram, implementation timeline, KPI dashboard mockup.

Discussed benchmark data sources and agreed to prioritize peer-reviewed healthcare studies over generic reports.

Decisions Made:

- Mario to create a roadmap diagram using Canva by 23/02/26.
- All members to ensure consistent Harvard citation format in their sections.
- Emir to compile all references into the unified References section.
- Alex to source additional healthcare AI case studies for the Feasibility section.

Action Items

Task	Assigned To	Deadline
Create Roadmap diagram	Mario	23/02/26
Add 3-5 additional academic citations	Alex	24/02/26
Standardize all citations to Harvard format	Everyone	24/02/26

Draft Implementation Roadmap Tables	Emir	25/02/26
Review and provide feedback on all sections	Everyone	25/02/26

Individual Contributions

Mario: Coordinated section integration, assigned citation tasks.
Emir: Volunteered to create tables, provided AI framework research.
Alex: Identified citation gaps, offered to research additional case studies.
Ilja: Reviewed Executive Summary for consistency with the main body.
Vasileios: Cross-checked Problem Analysis data against scenario requirements.

Meeting 4

Date: 28/02/26

Time: 3pm

Platform: WhatsApp

Attendees: Mario, Emir, Alex, Ilja, Vasileios

Absent: None

Agenda

- Final draft review and submission preparation.

Discussion Summary:

Agenda includes reviewing completed sections, final editing, consistency checks, and submission prep.

Discussed what we need to review (all report sections) and quality checks.

Set clear deadlines for proofreading (01/03/26) and final submission (03/03/26).

Decisions Made:

- Distributed editing/reviewing tasks across all team members.
- Decided on deadlines for proofreading and final submission.
- Decided on the next meeting for presentation preparation.

Action Items

Task	Assigned To	Deadline
Proofreading & quality checks	Everyone	1/03/26

Individual Contributions

Mario: Led meeting.

Emir: Took meeting notes.

Alex: Updated meeting log.

Ilja: Grammar checks.

Vasileios: Layout checks.

Meeting 5

Date: 01/03/26

Time: 12pm

Location: DCU Library

Attendees: Mario, Emir, Alex, Ilja, Vasileios

Absent: None

Agenda

- Presentation slide deck design and content
- Assign speaking roles for 5-minute oral presentation
- Rehearse presentation timing
- Final report submission checklist review

Discussion Summary:

Reviewed draft slide deck created by Mario – 12 slides covering problem, solution, feasibility, and roadmap.

Discussed presentation flow and agreed on 5-slide core structure: (1) Problem Overview, (2) AI Solution Architecture, (3) Implementation Phases, (4) ROI & Feasibility, (5) Recommendation.

Allocated speaking time: ~1 minute per person to fit within the 5-minute limit.

Assigned specific slides to each member based on their report sections for consistency.

Conducted practice run – initial timing was 6:30 minutes, agreed to condense content and remove 2 detail slides.

Decisions Made:

- Each member speaks for ~50-60 seconds on their assigned section.
- Mario opens (Problem), Emir covers Solution, Alex presents Feasibility, Vasileios discusses Ethics, Ilja concludes with Roadmap/Recommendation.
- Schedule final presentation rehearsal for 02/03/26 evening via Zoom.

Action Items

Task	Assigned To	Deadline
Finalize slide deck	Mario	02/03/26
Prepare opening statement (Problem)	Alex	02/03/26
Prepare Feasibility section speaking notes	Emir	02/03/26
Prepare Ethics section speaking notes	Vasileios	02/03/26
Prepare Roadmap/Conclusion speaking notes	Ilja	02/03/26
Conduct final presentation rehearsal (Zoom)	Everyone	02/03/26
Submit final report PDF via Loop	Emir	01/03/26

Individual Contributions

Mario: Led presentation is he planning, assigned speaking roles, coordinated group meetings.

Emir: Created and produced the final slide deck, provided visual design direction, assigned speaking roles to team members.

Alex: Timed practice run, identified areas needing condensing.

Ilja: Volunteered to handle conclusion/recommendation slide, suggested removing 2 detail slides.

Vasileios: Reviewed slide content for alignment with report, suggested accessibility improvements (contrast, font size).

Conflict Resolution Strategy

In the event of disagreements or conflicts within the team, the following protocol will be followed:

1. Open discussion during team meetings to address concerns.
2. Democratic voting on contested decisions (majority rule).
3. Escalation to module coordinator if solution cannot be reached.
4. Regular check-ins to ensure all team members feel heard and valued.

GenAI Usage Declaration

This section documents the team's use of Generative AI tools in compliance with the Level 2/3 guidelines specified in the assignment brief.

Tools Used

- ChatGPT: Research, planning.
- Grammarly : Proof reading.
- Canva: Diagrams.

Declaration: All content in the final report and presentation was originally written by team members. AI tools were used only for the purposes stated above, and no AI-generated content was directly included in the submission.

Notes

- This log should be updated after each team meeting.
- All team members should review and approve meeting summaries.
- Keep detailed records of individual contributions for the Appendix.
- Maintain evidence of work distribution (e.g., document version history, commit logs).
- Ensure fair and equitable workload distribution throughout the project.

Appendix C: Presentation Slide Design Commentary

Design Approach:
We adopted a professional healthcare aesthetic with clean layouts and limited text per slide to ensure clarity during the 5-minute presentation.

Key Design Choices:

1. Color Scheme:
- Chose blue/white palette to convey trust and professionalism (appropriate for healthcare)
- Avoided red tones (associated with errors/danger in medical contexts)

2. Typography:
- Calibri fonts (12pt for body)

3. Visual Hierarchy:
- Used icons for each system component (Data, Analytics, Scheduling, Governance)
- Timeline visualization for 18-month roadmap
- KPI dashboard mockup on outcomes slide

4. Accessibility:
- High contrast ratios for readability
- Alt text planned for all images

Rationale:
These choices reflect best practices from the lecture notes on presentation design, balancing professionalism with engagement for a senior management audience.

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